Quantum Cross-Family and Partial-Family Prediction

Data-driven transfer with observable contracts, normalization and abstention

Thy Nguyen

29 October 2026
Mathematical methods manuscript, revised edition
DOI: 10.5281/zenodo.23783870

Abstract

Quantum families can share measurable predictive structure without sharing a Hilbert space, a Hamiltonian or a universal transition law. This paper formulates cross-family transfer at the level of normalized observations and target semantics. It develops component-wise eligibility, same-target aggregation, finite-size and clock normalization, partial transfer between light–matter and interacting-lattice models, and uncertainty-aware calibration. Trace-distance bounds, conditional-law transport and exact counterexamples identify what the transfer procedure can and cannot justify. The inference model learns from data; displayed physical equations define examples and constraints rather than supplying the hidden generator to the predictor.

Research lineage and mathematical status. This study develops a WPM mathematical extension with reference to the author's published WPM architecture [1] and foundational n/s scaling-spectrum framework [2]. The n/s paper concerns conditional moment-spectrum parameterization; the WPM paper provides the broader architectural reference. The results proved here are mathematical consequences of stated assumptions, usually applications of standard linear algebra, probability or optimization. The WPM contribution is their organization into an explicit workflow. Modeling constructions and applicability conditions are identified separately; no universal forecasting superiority is asserted.

Predictor inputs. The forecasting layer described here is data driven: it learns from observed histories and matured outcomes. It is not fed the system's true governing equation or unobserved future states at issuance. Physical equations used below define observables, explain a limiting example or specify a separately labeled comparator. Optional supplied-map inverse problems and physics-informed variants must be declared separately; they are not evidence that the default forecast knows the generator.

1 Transfer an observable, not an unnamed quantum state

Let family f have state space $\mathcal{H}_f$, state ρ_f, observed channels X_f and requested target Y_f^T. A density matrix from an N-spin chain cannot be averaged directly with one from a truncated cavity unless an explicit common state embedding or channel is defined. Matching matrix dimensions by padding is not a physical equivalence.

A more general transfer route uses common observable features $Z_f = N_{S,f}(X_f, C_f)$ and a common target $U_f = N_{T,f}(Y_f^T)$. The learned forecasting map $g_T(Z_{\le t}, P)$ is then shared only where the

normalized conditional prediction problem is compatible. This supports transfer between different physical families without claiming that their wavefunctions or equations have become identical.

Every output retains the observation contract: preparation and measurement scheme, sample clock, units, known settings, missing channels, region definitions and horizon. A supplied Hamiltonian for an analytical illustration is different from a learned model receiving it at inference. If an implementation computes a feature from an unobserved true state or exact bath rate, it is a privileged-state variant and must not be described as an observation-only forecast.

2 Dual normalization and the fixed physical endpoint

System normalization aligns coordinates and context. Examples include x/L, s_x/L, density mass on a common grid, g/ω_c, U/J, detuning ratios and bath-memory clock λq. Target normalization aligns the meaning of the future observable or event.

If a physical horizon is q and sampling is Δ_f, the horizon is $P_f = q/\Delta_f$ under that convention. A coupling clock $g\sqrt{\bar{n}+1}q$ may help represent the query, but it does not change q. Unknown clock parameters add uncertainty. If cross-family transfer instead defines equal dimensionless duration, explicitly state that the physical endpoints differ by family.

Worked example 2.1 (Target scale is not just feature scale)**.** For a kicked rotor, use a declared reference scale $D_{\mathrm{ref}}(K)$ and transport increment $R = [\mathrm{Var}\, p(t+h) - \mathrm{Var}\, p(t)]/[D_{\mathrm{ref}}(K)h]$. A threshold $R > b$ can express relative transport across K. Standardizing K among the input features does not turn a fixed absolute-width threshold into that same event. Conversely, changing the target to R is a new declared query, not a silent repair of the old one.

For a spatial density, interpolate probability masses conservatively or track remapping error. Density heights and site probabilities have different normalization weights. A region attached to a defect should be registered relative to the defect when the physical target is defect-local; a fixed laboratory region should not move merely to make source and destination look similar.

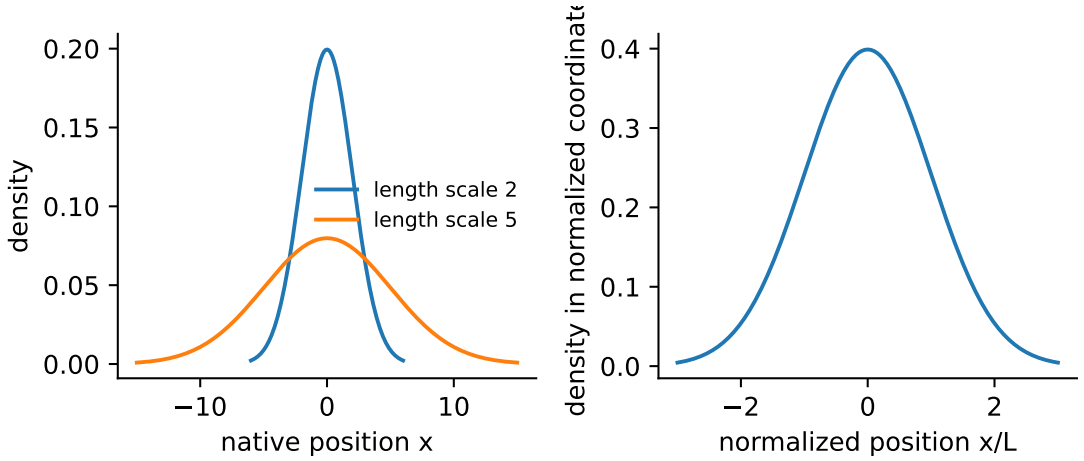


Figure 1: Two stipulated density profiles in different native length units become the same normalized mass density. The Jacobian factor is essential. Coordinate agreement alone does not establish the same future conditional law.

3 A sufficient condition for common predictive risk

Let source and destination normalized laws be $P_s(Z,U)$ and $P_d(Z,U)$. A common feature label does not imply a common conditional law.

Theorem 3.1 (Conditional-law transport)**.** *If $P_s(U \mid Z) = P_d(U \mid Z)$, P_d^Z is absolutely continuous with respect to P_s^Z, and loss is integrable, then for $w = dP_d^Z/dP_s^Z$,*

$$R_d(g) = \mathbb{E}_s[w(Z)\ell(U, g(Z))]. \tag{3.1}$$

Proof. Write destination risk as an iterated integral over Z and its conditional U distribution. Substitute the equal source conditional law and density-ratio representation of the destination marginal. This yields the weighted source expectation. □

The theorem is conditional on invariance, support coverage and a specified target. It does not derive those assumptions from similar pocket geometry. An unseen coupling interval can violate support coverage; an environment change can alter the conditional law; and a different measurement protocol can change the available features.

On a bounded scalar target, an alternative perturbation argument can use a coupling with $|U_d - U_s| \le \delta_T$, $\|Z_d - Z_s\| \le \delta_S$, a K-Lipschitz g and an L-Lipschitz loss. The risk difference is then bounded by $L(\delta_T + K\delta_S)$. Each bound must be justified for that domain. Squared loss needs bounded predictions/targets or another moment argument to obtain global Lipschitz control.

4 Partial-family eligibility before any mixture

Index candidate family f, target component c and resolution pocket k. Define

$$I_{fck} = I_{fc}^{\text{observable}} I_{fck}^{\text{semantic}} I_{fck}^{\text{qualified}}. \tag{4.1}$$

The first factor checks that the required channel is available and meaningful, the second that it predicts the same target in compatible units, and the third that the policy permits use at the requested horizon and context. An eligibility mask is not a correctness probability.

Partial eligibility permits a family's density-geometry contribution while withholding unsupported coherence, environmental-memory or high-frequency contributions. The mask acts on forecasting candidates. It must not remove counter-rotating terms from the real physical system merely because a simpler predictor cannot represent them. Missing physical dynamics and withheld forecast components are different concepts.

For alternative estimates of one U, use $\widehat{U}_c = \sum_{f,k} \pi_{fck} \widehat{U}_{fck}$ over eligible entries with nonnegative weights summing to one. If $Y = \sum_c Y_c$ is a required additive target, first form each component forecast and then assemble the sum. A missing required component causes partial reporting, a valid fallback or abstention. It must not disappear through renormalization of the remaining components.

Worked example 4.1 (Disjoint regions and coherent amplitudes)**.** For disjoint regions R_1, R_2, occupation of their union is $B_1 + B_2$. Predicting only B_1 is not a total-union forecast. In contrast, two coherent amplitudes overlapping in one region give an interference cross term; adding separate intensities is not the union rule. The observation/target definition decides which arithmetic is applicable.

5 Quantum constraints and a trace-distance proof trail

Theorem 5.1 (Mixture of compatible density forecasts)**.** *If all ρ_i are positive semidefinite unit-trace operators on the same Hilbert space and $\pi_i \geq 0$, $\sum_i \pi_i = 1$, then $\rho = \sum_i \pi_i \rho_i$ is a density operator. For a fixed measurement effect E, its predicted probability is* $\operatorname{tr}(E\rho) = \sum_i \pi_i \operatorname{tr}(E\rho_i)$.

Proof. For any vector v, $v^\dagger \rho v = \sum_i \pi_i v^\dagger \rho_i v \geq 0$. Trace is linear and equals one. The probability equality is trace linearity. □

This mixture represents alternative state descriptions; it is not the entangled joint state of several simultaneously present systems. Different Hilbert spaces require a lawful mapping or observable-level combination instead.

Theorem 5.2 (Occupation sensitivity to state error)**.** *For two density matrices on the same space let $D = \frac{1}{2} \|\rho - \sigma\|_1$. Every effect $0 \preceq E \preceq I$ satisfies $|\operatorname{tr}[E(\rho - \sigma)]| \leq D$.*

Proof. Let $\Delta = \rho - \sigma = \Delta_+ - \Delta_-$ be its positive/negative decomposition. Since $\operatorname{tr}\Delta = 0$, both positive traces equal D. The effect bounds give $0 \leq \operatorname{tr}(E\Delta_\pm) \leq D$, so their difference lies between -D and D. □

Thus an occupation threshold label is stable if the true occupation is farther than D from the threshold. Near the boundary, even a good state approximation can change the event label. For general bounded observables, $|\operatorname{tr}[O(\rho - \sigma)]| \leq \|O\|_\infty \|\rho - \sigma\|_1$. These inequalities concern comparable states, not arbitrary transfer between unmatched quantum models.

6 Light–matter partial transfer: JC and full Rabi

The Jaynes–Cummings rotating-wave model and full Rabi model share atom/cavity observables but differ in counter-rotating dynamics. The JC excitation operator $N = a^\dagger a + \sigma_+\sigma_-$ commutes with the JC Hamiltonian. It does not generally commute with the full Rabi Hamiltonian. Therefore a forecast based on exact excitation conservation is not a universally transferable component.

Conservation trail. Using $[a^\dagger a, a] = -a$ and $[\sigma_+\sigma_-, \sigma_+] = \sigma_+$, the two contributions cancel in $[N, a\sigma_+]$. They also cancel for $a^\dagger\sigma_-$. For the counter-rotating term $a^\dagger\sigma_+$ they add rather than cancel. Hence the conservation assertion depends on the coupling model. □

Common observed excitation, photon-number geometry and detuning can still support empirical transfer. Retain the source's qualified same-target candidates while excluding unsupported conservation-based forecasts. At ultrastrong coupling, keep extra observed context or abstain if the source mapping is inadequate; do not relabel the full system as JC to force compatibility.

A coupling-clock normalization can make oscillatory histories comparable locally. For a resonant two-state block, equal $g\sqrt{n+1}q$ gives equal ideal excitation phase. A superposition of photon numbers, detuning or counter-rotating terms breaks that simple one-clock description. The data-driven model must qualify its use rather than inheriting exactness from the limiting example.

7 Interacting lattices: common geometry and different interactions

The Bose–Hubbard onsite term $Un_i(n_i - 1)/2$ and a hard-core/spinless nearest-neighbor term Vn_in_{i+1} are not identical. Shared density mass, centroid, width, occupation, normalized potential profile and fluctuation descriptors can define an observable interface. Transfer of those descriptors does not prove equality of all correlations or transition laws.

In one-dimensional hard-core settings, a Jordan–Wigner transformation connects local density operators to spinless-fermion densities, with boundary/parity subtleties. This special mathematical relation is not a claim that a finite-U Bose–Hubbard system equals every spinless chain. Partial-family routing can use a common density target while withholding source-specific double-occupancy or coherence predictions not supported in the destination.

Worked example 7.1 (An exactly shared single-particle sector)**.** On two sites with one particle and identical hopping J, both an onsite-interacting bosonic model and a nearest-neighbor-interacting spinless model have vanishing interaction energy in that sector. Both reduce, up to supplied onsite potentials, to a two-state hopping problem. This provides a restricted exact transfer example. With two particles, the allowed states and interaction energies can differ, so the argument no longer applies.

If separate predicted region masses violate $0 \leq B_i \leq 1$ or $\sum_i B_i \leq 1$, enforce a joint feasible model or report a constraint repair. Such a repair restores algebraic consistency but does not guarantee transport of calibration.

8 Within-family crossings: size, disorder and drive location

Within-family transfer can fail even when Hamiltonian form is unchanged. Chain length changes finite-size behavior; moving a disorder region changes the observation geometry; a withheld coupling interval can alter predictability. Use physically meaningful coordinates before deciding whether adaptation is needed.

For an Ising scaling ansatz, $x_h = (h/J - 1)N^{1/\nu}$ and $x_M = M_z N^{\beta/\nu}$ encode a specified critical regime. They are not a theorem about arbitrary driven finite chains. A raw-magnetization threshold may need a different target map from a scaled-magnetization threshold. State which physical query is preserved.

For a localized potential or disorder window, relative coordinates $(x - x_{\text{defect}})/L$ can remove a mere translation when the target is defect-relative. They can be wrong for a laboratory-fixed target. Likewise, adding more dimensionless disorder features is not guaranteed to repair a changed conditional law. If representation corrections remain inadequate, a separate, small target calibration set can support probability-scale adaptation, with a new qualification check.

9 Open-system crossings and target semantics

A Markovian Lindblad family and a memory-bearing reservoir may share measured populations while having different future conditional laws. A population revival is not a universal definition of information backflow. If the target is trace-distance backflow, source and destination must use a comparable state-pair and observation protocol. A nonnegative-rate Markovian source alone may provide no positive examples for that event.

For an amplitude-damping family, bath-memory ratio λ/γ_0 and duration λq can organize context. If these parameters are unknown at inference, their estimates carry uncertainty or must be omitted. A correct dimensional representation cannot identify an unmeasured bath uniquely. Additional temporal history may reveal memory, but a finite observed population sequence can still admit multiple environments.

10 Floquet phase and transport crossings

Driven systems require drive-phase alignment when sampling within a period. Two identical phase-free feature vectors can correspond to different positions in a drive cycle and different near-future trajectories. Phase-tagging or stroboscopic sampling addresses that representation issue; it does not establish a universal shared response across drive amplitudes.

For kicked-rotor transport, distinguish a source-normalized relative diffusion query from a destination absolute-width query. The latter may be physically meaningful, but it is not the same label. Zero-shot transfer means no destination fitting, routing selection or calibration using its outcomes; using known instrument settings or a predeclared physical unit conversion is a different information category.

11 Calibration, mode choice and evidence scope

Fixed, Multi and Auto remain legitimate after eligibility routing. Similar pocket shapes are evidence for a representation hypothesis, not proof that source probabilities are calibrated at destination. A monotone calibration map can change Brier while preserving ranking; it cannot manufacture positive-event discrimination in a destination subset with no positive examples.

Evidence statement. Archived quantum reports describe selected lower-Brier data-driven transfer forecasts against their specified Fixed or conventional observation-based references, with failures in other regimes [4, 5]. The available descriptions do not certify those references as generator-equation-fed models. A smaller Brier score than a misspecified equation-informed predictor is mathematically possible, but a correctly specified same-information conditional-probability oracle remains optimal. Baseline inputs, target semantics and information cutoffs must accompany any stronger comparison.

The mathematical workflow is therefore: define the observable contract, normalize coordinates and targets, retain compatible pockets, mask unsupported contributions, choose a qualified forecast policy, and calibrate only where the assumptions permit. If no required contribution remains, abstain. Inverse recovery of a new physical relationship is a separate task requiring identifiability; successful predictive transfer alone does not derive a new Hamiltonian.

12 Application worksheet and exercises

Suppose two quantum families estimate the same normalized future region occupation with forecasts 0.2 and 0.6. Base weights are 3 and 1. If both are eligible, the mixture is $(3\times0.2+0.6)/4 = 0.3$. If the second is unqualified, the routed forecast is 0.2, not 0.15. If instead the two numbers estimate required occupations of disjoint regions, their union estimate is 0.8 and it is incomplete when either required term is withheld.

1. Specify family, measured channels, target, duration and observation protocol.

2. Record N_S, N_T, domain masks and uncertainty of fitted settings.
3. Check channel availability, semantic identity and qualification separately for each candidate component and pocket.
4. Preserve individual flow slices, positions, existence and regime identities.
5. Aggregate same-target alternatives; assemble required components with their dependence and physical constraints.
6. State whether the output is zero-shot, target-calibrated or physics-informed. Keep these labels distinct.

Exercise 12.1. Can a cavity density matrix and a spin-chain density matrix be mixed merely because both were stored as square arrays?

Solution. No. A shared state space and physical mapping are needed. Common observable forecasts may be combined under a separate target contract.

Exercise 12.2. If state trace distance is at most 0.03 and the true region occupation is 0.70, is a threshold 0.65 stable?

Solution. Yes under the stated comparable-state bound: predicted occupation is at least 0.67. At a threshold 0.69, stability is not certified.

Scope of the mathematical exposition

Unless explicitly identified as a sourced evidence statement, numerical values are constructed arithmetic examples. Application requires specifying the observation process, fitting unknown quantities, and checking reliability for the intended population and horizon. This study and its quantum companion (Papers 7 and 8 of the series) share the archival DOI 10.5281/zenodo.23783870.

References

[1] T. Nguyen. *The WPM Framework: A Generalized Mathematical Architecture for Cross-Scale, Spatial, and Dynamical Prediction*. Zenodo, 2026. doi:10.5281/zenodo.23248983.

[2] T. Nguyen. *An n/s Scaling-Spectrum Framework for Generalized Moment Scaling: Curvature, Signed-Moment Asymmetry, and Prospective Moment-Order Validation*. Zenodo, 2026. doi:10.5281/zenodo.23111783.

[3] T. Gneiting and A. E. Raftery. Strictly proper scoring rules, prediction, and estimation. *Journal of the American Statistical Association* 102(477), 359–378, 2007. doi:10.1198/016214506000001437.

[4] T. Nguyen. *The Multi-Lens W-P-m Framework: A Resolution-Spectrum Extension for Structural Forecasting Across Chaotic and Quantum Dynamics*. Archive version 24: dual normalization and non-Markovian extension. October 2026. Referenced for method provenance and scoped baseline statements, not as a proof of universal superiority.

[5] T. Nguyen. *Target-Conditioned Temporal Resolution in Quantum Forecasting: Location, Scale, Structural Context, and Foreseeable Boundary Risk*. Version 2, October 2026. Archived manuscript.

[6] M. Sugiyama, M. Krauledat and K.-R. Müller. Covariate shift adaptation by importance weighted cross validation. *JMLR* 8, 985–1005, 2007. https://jmlr.org/papers/v8/sugiyama07a.html.

[7] QuTiP developers. *Using tensor products and partial traces*. Official documentation. https://qutip.org/docs/4.6/guide/guide-tensor.html.